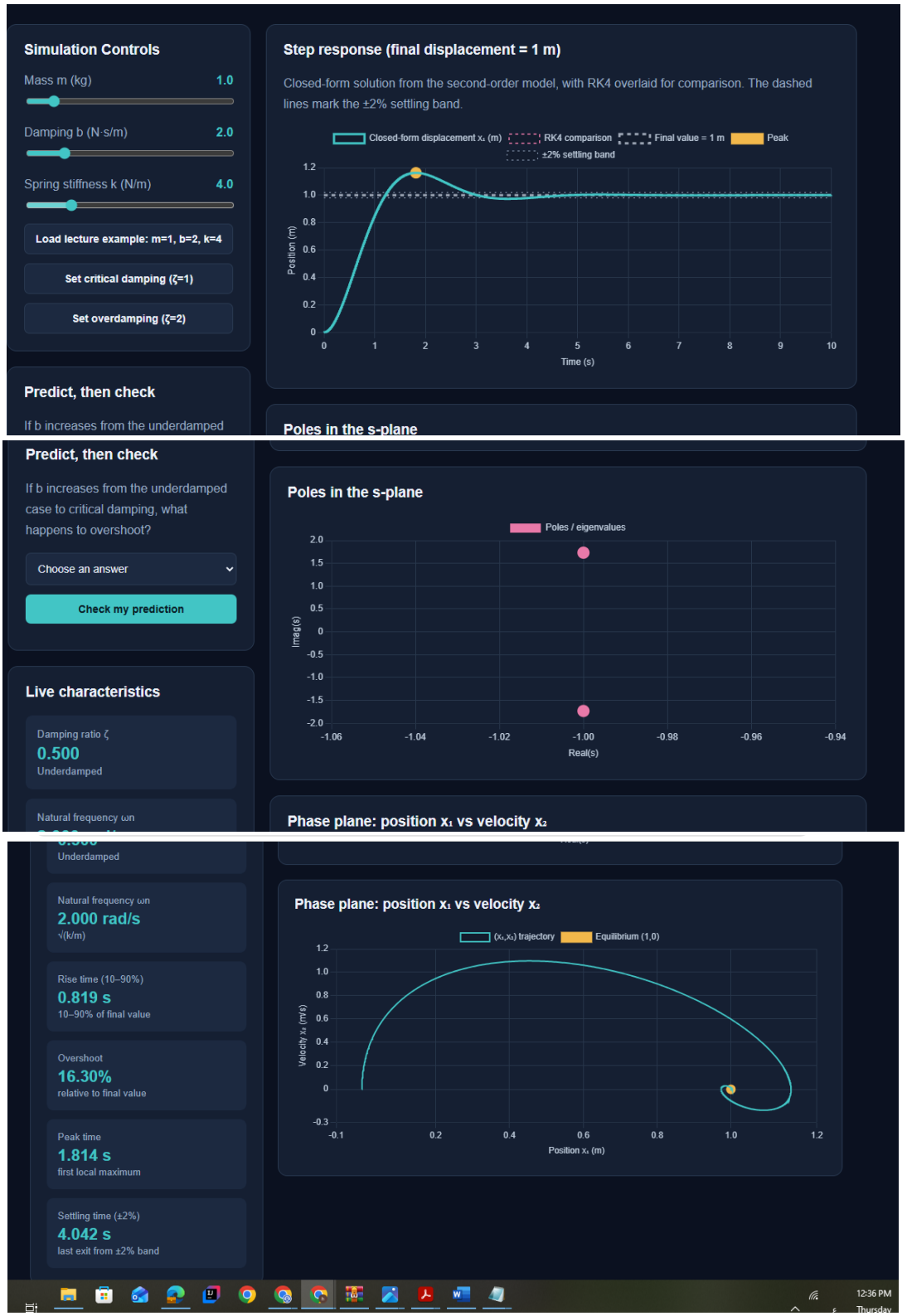
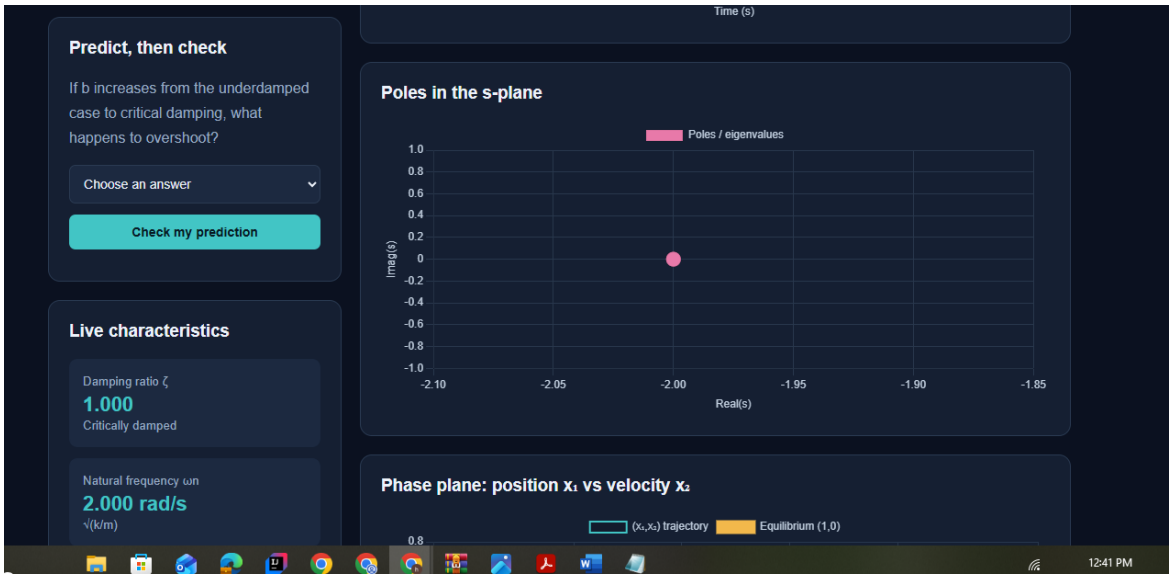
Module 3 Test Cases

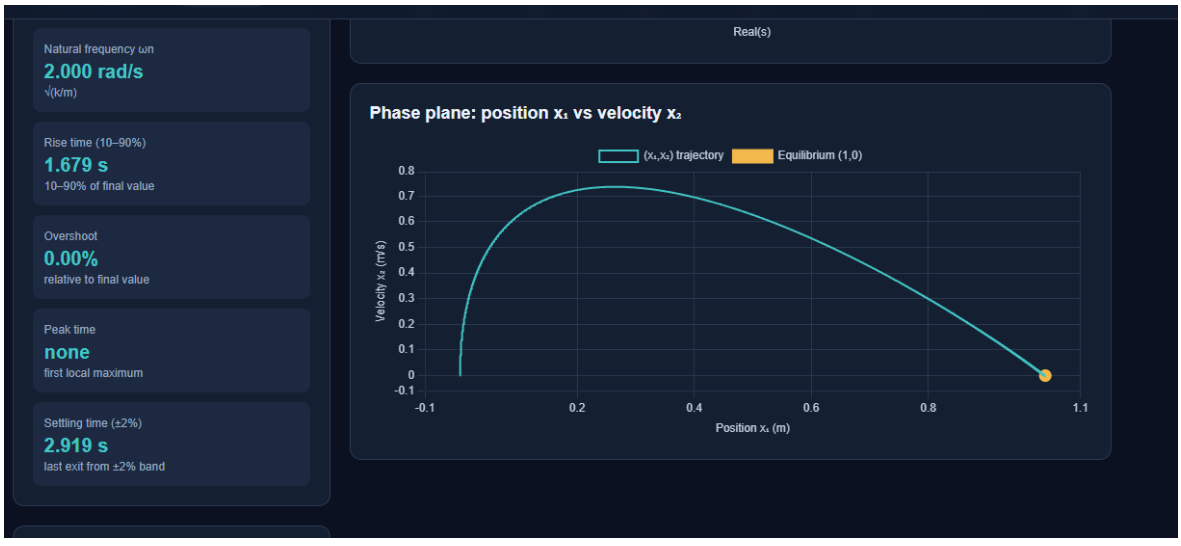
TC01 — Underdamped Response (ζ = 0.5)



TC02 — Critical Damping (ζ = 1)



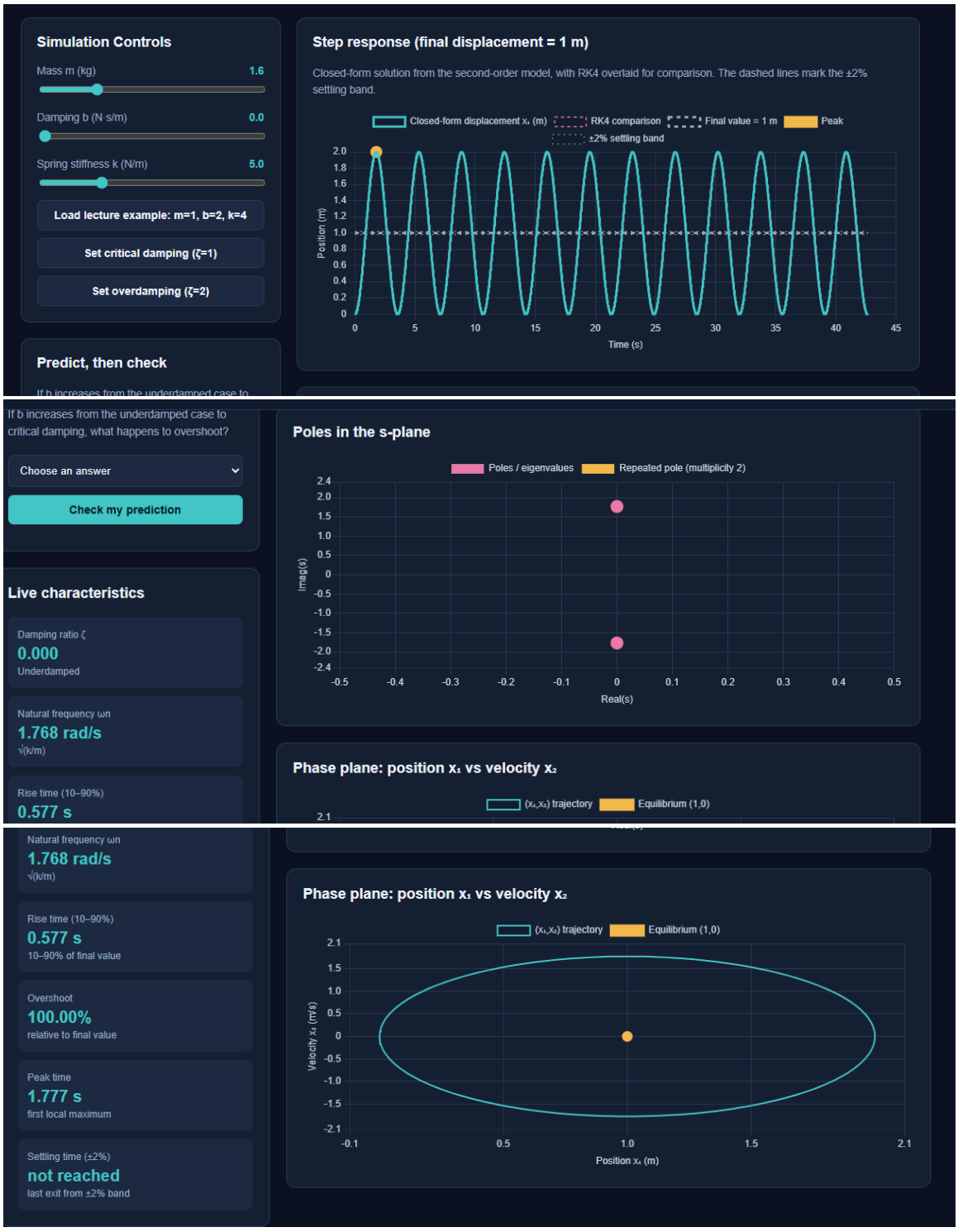




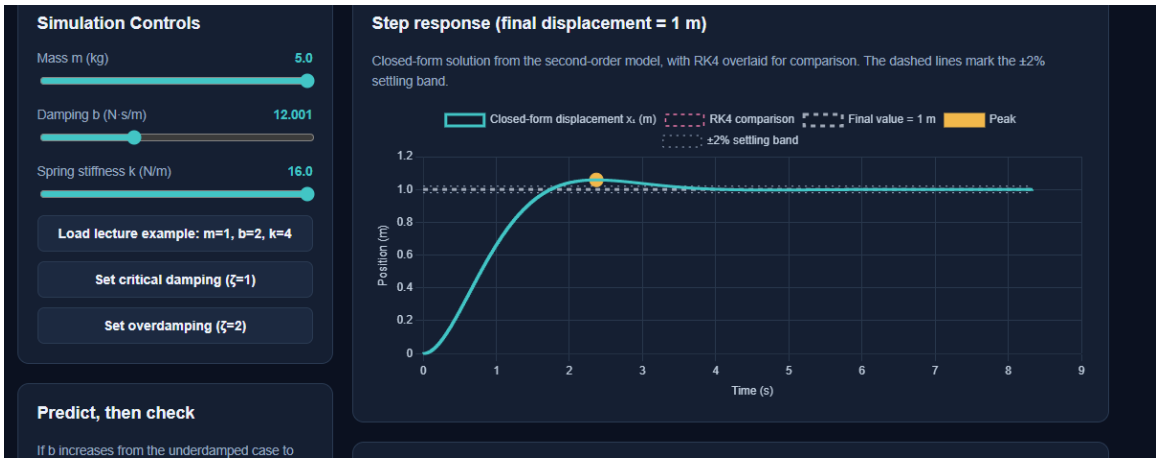
TC03 — Overdamped Response (ζ = 2)



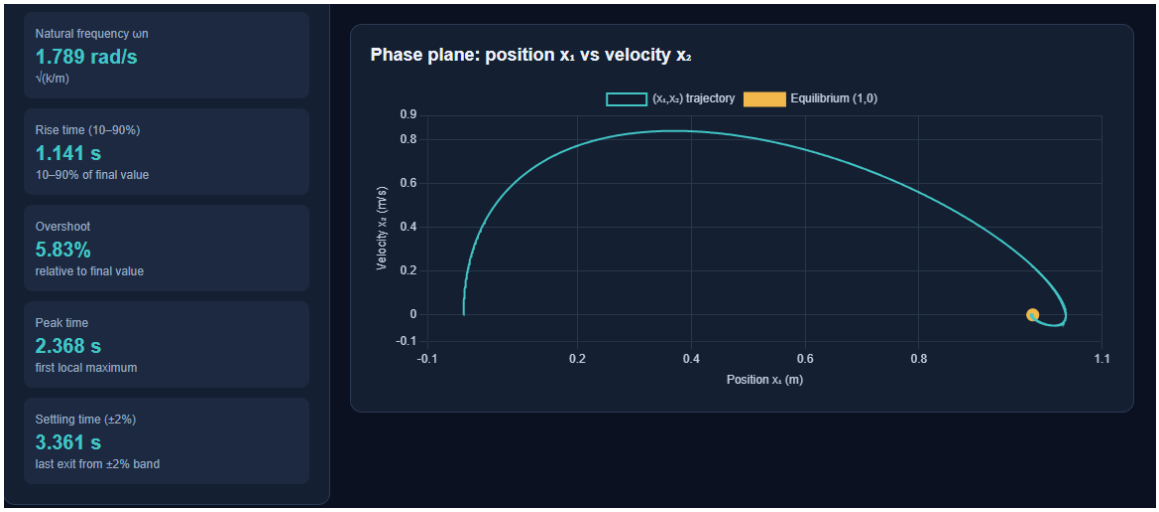
TC05 — Zero Damping (ζ = 0)



TC06 — High Mass & Stiffness Parameters





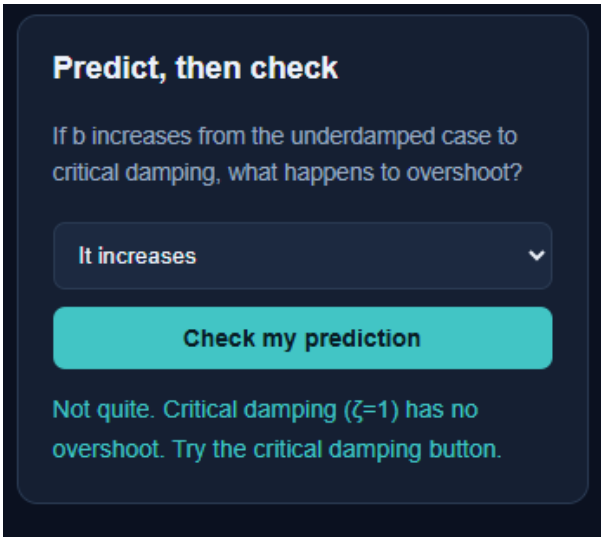


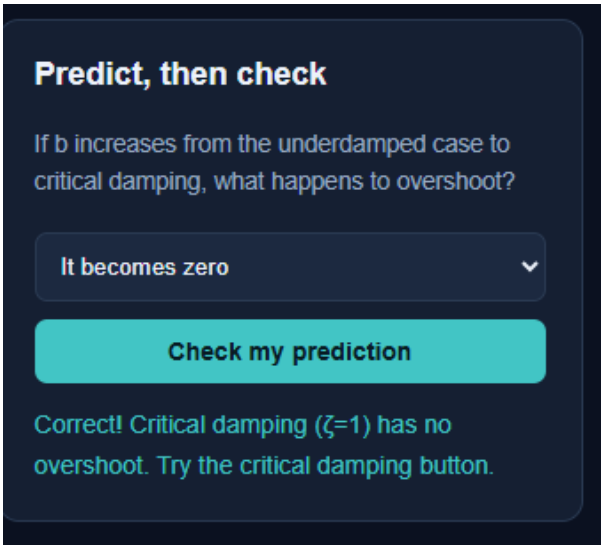


TC07 — Poles & State-Space Matrices



TC08 — Predict, Then Check





TC09 — Lecture Validation Checks

Validation checks

Check	Expected	Your value	Result
4. ζ=0.5 overshoot	16.3%	16.30%	✓ PASS
5. ζ=1 overshoot	0%	0.000%	✓ PASS
6. m=1, b=2, k=4 eigenvalues	−1 ± j1.732	-1.000 ± j1.732	✓ PASS